SMB Surface Mount Schottky Barrier Rectifier

Features

- For surface mounted applications
- High forward surge current capability
- Low power loss,high efficiency
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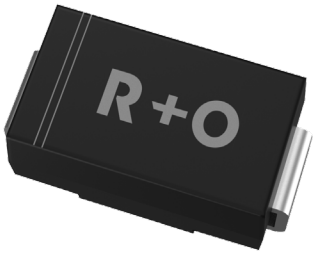
- Meets MSL level 1, per J-STD-020, LF maximum peak of 260°C

Reverse Voltage
20-100 V
Forward Current
10 Ampere

Applications

For use in low voltage high frequency inverters, free wheeling,DC/DC converters, and polarity protection applications.

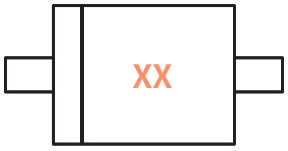
DO-214AA(SMB)

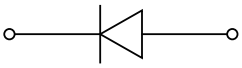


Mechanical Data

- Case: DO-214AA(SMB)
  Molding compound meets UL 94V-0 flammability rating, RoHS-compliant, halogen-free
- Terminals: Solder plated, solderable per MIL-STD-750,Method 2026
- Polarity: Cathode line denotes the cathode end

Function Diagram





Maximum Ratings (Ta=25°C Unless otherwise specified)

PARAMETER	SYMBOL	UNIT	SK102B	SK104B	SK1045B	SK106B	SK108B	SK1010B	SK1015B	SK1020B
Device marking code			SK102B	SK104B	SK1045B	SK106B	SK108B	SK1010B	SK1015B	SK1020B
Maximum Repetitive Peak Reverse Voltage	V_{RRM}	V	20	40	45	60	80	100	150	200
Maximum RMS Voltage	V_{RMS}	V	14	28	32	42	56	70	105	140
Maximum DC blocking Voltage	V_{DC}	V	20	40	45	60	80	100	150	200
Maximum Average Forward Rectified Current @60Hz sinewave, Resistance load,TL (Fig.1)	$I_{F(AV)}$	A	10							
Non-repetitive Peak Forward Surge Current @ t=8.3ms Half-sine wave	I_{FSM}	A	250							
Storage temperature	T_{stg}	°C	-55 ~ +150							
Junction temperature	T_j	°C	-55 ~ +150							
Typical Thermal Resistance	$R_{\theta J-A}$	°C /W	55							
	$R_{\theta J-L}$	°C /W	15							

Electrical Characteristics (Ta=25°C Unless otherwise noted)

PARAMETER	TEST CONDITIONS	SYMBOL	UNIT	SK102B	SK104B	SK1045B	SK106B	SK108B	SK1010B	SK1015B	SK1020B
Maximum instantaneous forward voltage	I_F=10A	V_F	V	0.55			0.70		0.85	0.92	
Maximum DC reverse currentat rated DC blocking voltage	V_R=V_{DC}, T_A=25°C	I_{R1}	mA	0.1				0.02			
	V_R=V_{DC}, T_A=100°C	I_{R2}		5.0				2.0			
Typical junction capacitance	4.0V DC,1MHz	C_J	pF	400							

Ratings And Characteristics Curves (Ta=25°C Unless otherwise specified)

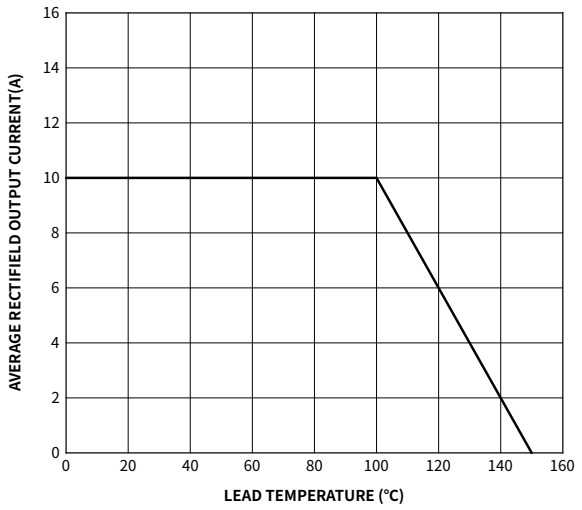


Fig.1 Forward Current Derating Curve

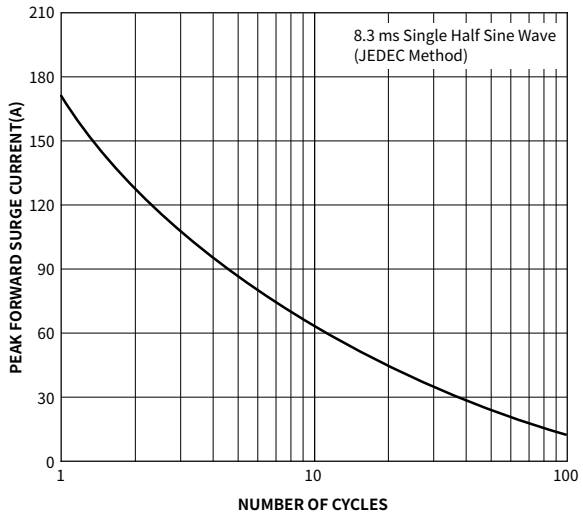


Fig.2 Maximum Non-Repetitive Peak Forward Surge Current

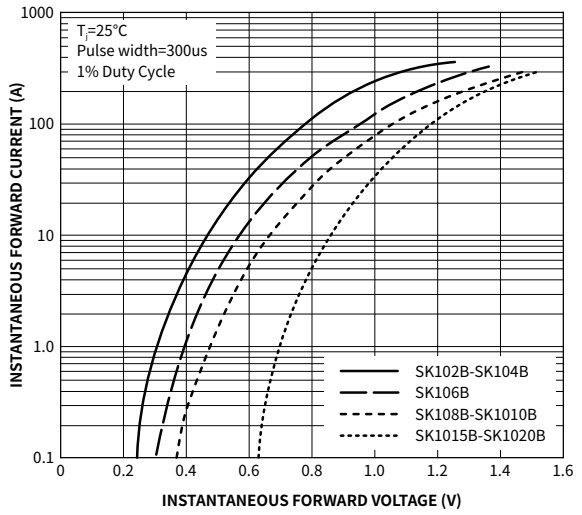


Fig.3 Typical Forward Voltage

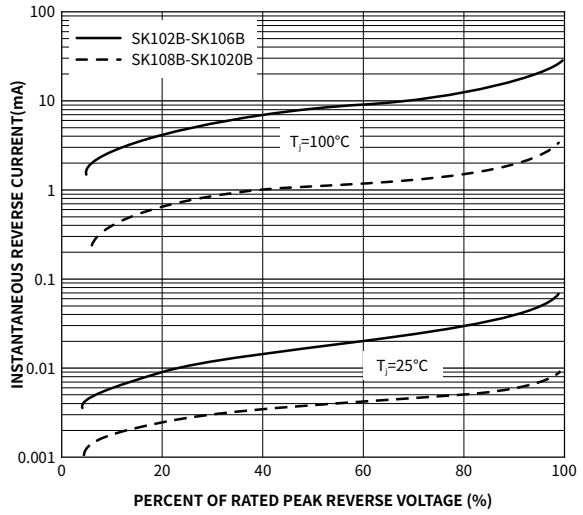
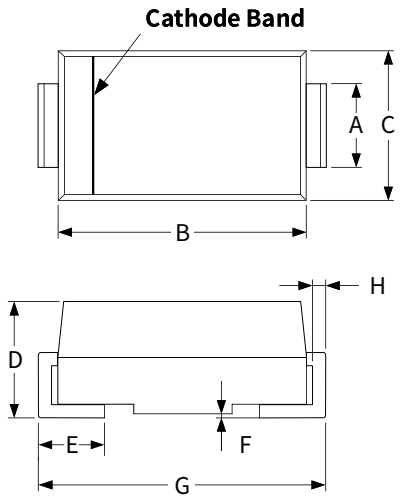


Fig.4 Typical Reverse Characteristics

Ordering Information

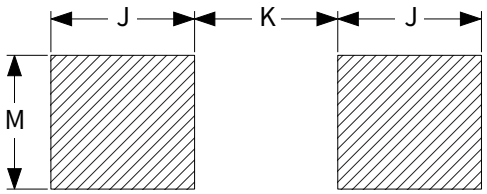
PACKAGE	PACKAGE CODE	UNIT WEIGHT(g)	REEL(pcs)	BOX(pcs)	CARTON(pcs)	DELIVERY MODE
SMB	R3	0.098	3000	6000	48000	13''

Package Outline Dimensions (SMB/DO-214AA)



Symbol	Dimensions			
	Millimeters		Inches	
	Min.	Max.	Min.	Max.
A	1.85	2.21	0.073	0.087
B	4.25	4.85	0.167	0.191
C	3.30	3.94	0.130	0.155
D	2.15	2.65	0.085	0.104
E	0.75	1.52	0.030	0.060
F	-	0.203	-	0.008
G	5.08	5.59	0.200	0.220
H	0.15	0.31	0.006	0.012

Suggested Pad Layout



Symbol	Dimensions			
	Millimeters		Inches	
	Min.	Max.	Min.	Max.
M	2.26	-	0.089	-
J	2.10	-	0.085	-
K	-	2.74	-	0.107